

Austin Prairie

Final Assessment Report Submission

Case: Triage Protocol

Executive Summary

Alert received from user computer concerning possible credential theft with suspicious activity outbound to foreign IP 158.159.10.22 through port 1443. Upon further investigation we ran process explorer to search for malicious files. Upon investigation, we discovered a process called MarketingTool.exe. Running its hash against virus total revealed 32/77 reported it as a malicious program, most pointing towards it being a trojan. The viruses hash is

add2fd5a5b8a144235ef63c3a0ae9036a1f54befde31b701e195d5c30bb6b5b6. The process when investigating further reveals that it has the name "malware.py" indicating that the likely attacker named it literally malware and the extension points to it being a python script that runs on startup. The behavior of the malware was to periodically take a screenshot of what the user was doing as well as creating a text file then sending it to the malicious ip address listed above. It would cache its work in a folder under C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\cache.

It would then open tasklist.exe through the created persistence mechanism. Further malware investigation would show that it was created by "Redevil@2024" with the config data value of 7ab85, giving itself permissions through the registry keys on the machine. When opening autorun you can see that both marketingtools.exe and the config data were there to run whenever the computer boots up. The dump log indicates that the target process is lsass.exe and the method of choice is memory scraping. To fix the machine, we sever the connection to the malicious host, kill the process of marketingdata.exe, remove the program and run an antivirus scan to destroy any unwanted fingerprints left and lastly we go into the registry keys to remove and reset anything that the malicious program tampered with.

Findings and Analysis

Finding	Finding Details	Description
Malicious File	MarketingTool.exe	The malicious file that came back as a trojan
Suspicious IP	158.159.10.22	Malicious IP that host machine was connected to
Company name	Redevil@2024	Company associated with the creation of the malware and possible implementation on machine
Virus hash	add2fd5a5b8a144235ef63c3a0ae9036a1f54befde31b701e195d5c30bb6b5b6	Virus signature used to identify
Dump file location	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\	Where the malware would drop its activity cache

Tools and Technologies Used

- Autorun – a utility from Microsoft Sysinternals that shows and manages every program, service, DLL and displays that is going to run through Windows boot up
- Procmonitor – A real-time monitor tool from Windows Microsoft Sysinternals.
- Procxplorer – a task managing system from Microsoft Sysinternals that gives detailed insights into running processes
- TCPView – a realtime Windows utility from Microsoft that provides real-time monitoring of active TCP and UDP traffic and includes local and remote addresses and what the status of them are.

Investigation Process

1. We begin by investigating process explorer to search for any suspicious processes

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- We discover a program called MarketingTool.exe, which if watched, occasionally opens a powershell.exe child under the process.

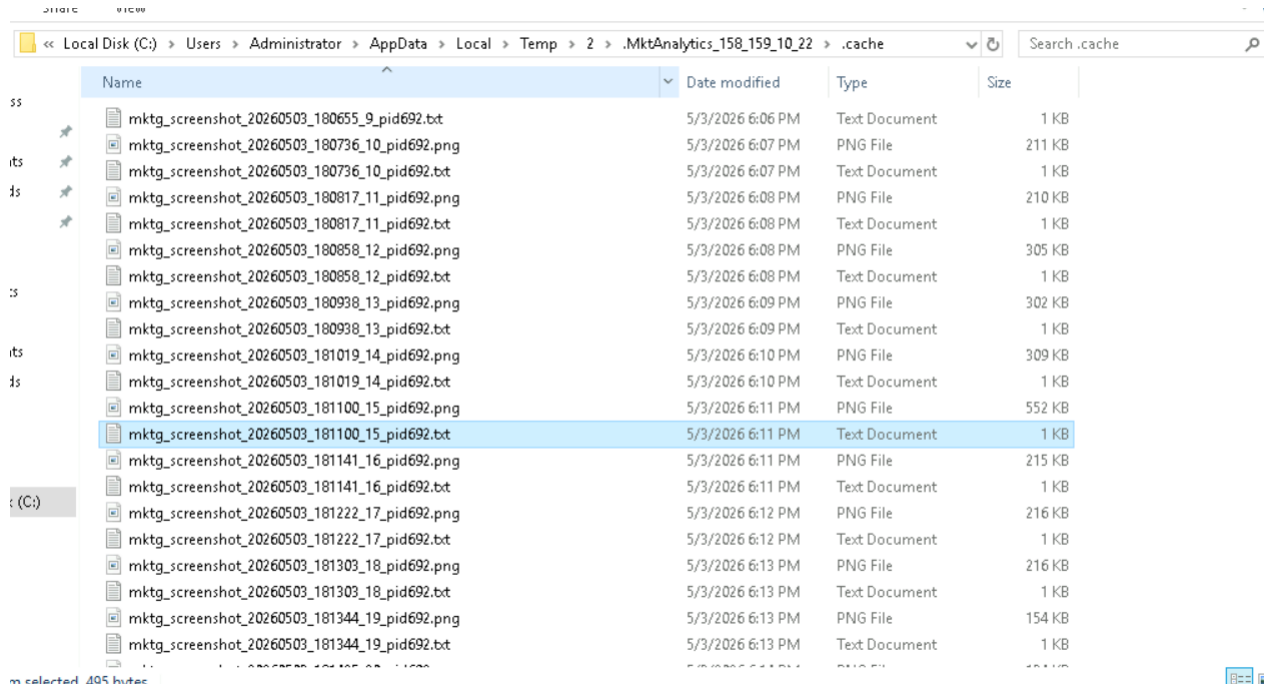
System Idle Process	99.17	56 K	8 K	0	
procexp64.exe	0.77	30,188 K	52,336 K	1484	Sysinternals Process Explorer Sysinternals - www.sysinter...
dwm.exe	< 0.01	17,248 K	72,712 K	4756	Desktop Window Manager Microsoft Corporation
Interrupts	< 0.01	0 K	0 K	n/a	Hardware Interrupts and DPCs
Procmon64.exe	< 0.01	72,020 K	41,936 K	3492	Process Monitor Sysinternals - www.sysinter...
System	< 0.01	192 K	160 K	4	
csrss.exe	< 0.01	2,480 K	5,688 K	744	
explorer.exe	< 0.01	31,644 K	113,300 K	4998	Windows Explorer Microsoft Corporation
MarketingTool.exe	< 0.01	1,264 K	6,124 K	5612	
dwm.exe	< 0.01	17,408 K	38,568 K	1020	Desktop Window Manager Microsoft Corporation
MarketingTool.exe	< 0.01	11,776 K	17,852 K	5680	
powershell.exe	< 0.01	2,160 K	4,984 K	476	

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- Taking this information, we navigate over to process monitor, filter by process MarketingTool (or process ID)

6/16/21 15:04:07 PM	MarketingTool...	5680	TCP Disconnect	Windows_Server:ec2:internal:49725 -> 158.159.10.22:1443	SUCCESS	Length: 0, segment: 0, connid: 0
6/16/21 15:06:03 PM	MarketingTool...	5680	ReadFile	C:\Windows\System32\mswsock.dll	SUCCESS	Offset: 392,192, Length: 2,560, I/O Flags: Non...
6/16/21 15:11:25 PM	MarketingTool...	5680	ReadFile	C:\Windows\System32\mswsock.dll	SUCCESS	Offset: 388,096, Length: 4,096, I/O Flags: Non...
6/16/21 15:17:33 PM	MarketingTool...	5680	ReadFile	C:\Windows\System32\mswsock.dll	SUCCESS	Offset: 384,000, Length: 4,096, I/O Flags: Non...
6/16/21 15:23:96 PM	MarketingTool...	5680	ReadFile	C:\Windows\System32\win2_32.dll	SUCCESS	Offset: 337,920, Length: 12,800, I/O Flags: No...
6/16/21 15:31:07 PM	MarketingTool...	5680	ReadFile	C:\Windows\System32\mswsock.dll	SUCCESS	Offset: 388,096, Length: 6,656, I/O Flags: Non...
6/16/21 15:41:12 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	SUCCESS	Desired Access: Read Attributes, Synchroniz...
6/16/21 15:41:58 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	SUCCESS	VolumeCreationTime: 4/15/2020 7:26:43 AM, ...
6/16/21 15:43:30 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	SUCCESS	CreationTime: 5/2/2026 6:16:16 PM, LastAcc...
6/16/21 15:44:07 PM	MarketingTool...	5680	QueryAllInfor...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	BUFFER OVERFL...	
6/16/21 15:44:26 PM	MarketingTool...	5680	QueryBasicInfor...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	SUCCESS	
6/16/21 15:44:58 PM	MarketingTool...	5680	QueryInformati...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	SUCCESS	
6/16/21 15:44:54 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\system\lvaas_log_20260502_181616_2_pid568...	SUCCESS	
6/16/21 15:45:25 PM	MarketingTool...	5680	RegOpenKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters	REPARSE	Desired Access: Read
6/16/21 15:45:48 PM	MarketingTool...	5680	RegOpenKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters	SUCCESS	Desired Access: Read
6/16/21 15:45:76 PM	MarketingTool...	5680	RegQueryValue	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters\Hostname	SUCCESS	Type: REG_SZ, Length: 30, Data: Windows_S...
6/16/21 15:45:87 PM	MarketingTool...	5680	RegCloseKey	HKLM\System\CurrentControlSet\Services\Tcpip\Parameters	SUCCESS	
6/16/21 15:53:55 PM	MarketingTool...	5680	TCP Reconnect	Windows_Server:ec2:internal:49725 -> 158.159.10.22:1443	SUCCESS	Length: 0, segment: 0, connid: 0
6/16/25 38:45:55 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	Desired Access: Generic Write, Read Attribute...
6/16/25 38:71:12 PM	MarketingTool...	5680	QueryInformati...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	VolumeCreationTime: 4/15/2020 7:26:43 AM, ...
6/16/25 38:71:50 PM	MarketingTool...	5680	QueryAllInfor...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	BUFFER OVERFL...	CreationTime: 5/2/2026 6:16:25 PM, LastAcc...
6/16/25 38:71:59 PM	MarketingTool...	5680	QueryBasicInfor...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	
6/16/25 38:72:12 PM	MarketingTool...	5680	QueryInformati...	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	
6/16/25 38:77:54 PM	MarketingTool...	5680	WriteFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	Offset: 0, Length: 308, Priority: Normal
6/16/25 38:79:37 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	
6/16/25 16:38:57 PM	MarketingTool...	5680	TCP Disconnect	Windows_Server:ec2:internal:49725 -> 158.159.10.22:1443	SUCCESS	Length: 0, segment: 0, connid: 0
6/16/26 16:41:28 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	Desired Access: Read Data/List Directory, Syn...
6/16/26 16:41:38 PM	MarketingTool...	5680	QueryDirectory	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	FileInformationClass: FileBothDirectoryInformati...
6/16/26 16:42:19 PM	MarketingTool...	5680	QueryDirectory	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	FileInformationClass: FileBothDirectoryInformati...
6/16/26 16:42:79 PM	MarketingTool...	5680	QueryDirectory	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	NO MORE FILES	FileInformationClass: FileBothDirectoryInformati...
6/16/26 16:43:07 PM	MarketingTool...	5680	CreateFile	C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22\config\marketing_report_20260502_181625_99...	SUCCESS	
6/16/23 45:48:75 PM	MarketingTool...	5680	CreateFile	C:\Windows\System32\powershell.exe	NAME NOT FOUND	Desired Access: Read Attributes, Disposition: ...
6/16/23 45:59:40 PM	MarketingTool...	5680	CreateFile	C:\Windows\System32\powershell.exe	NAME NOT FOUND	Desired Access: Read Attributes, Disposition: ...
6/16/23 45:57:01 PM	MarketingTool...	5680	CreateFile	C:\Windows\System32\powershell.exe	NAME NOT FOUND	Desired Access: Read Attributes, Disposition: ...
6/16/23 45:57:01 PM	MarketingTool...	5680	CreateFile	C:\Windows\System32\powershell.exe	NAME NOT FOUND	Desired Access: Read Attributes, Disposition: ...

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- We then see all the operations and registry keys that the program was changing, where it was creating logs, where the temp files were.
- Upon investigating the temp logs we find the it is stored in C:\Users\Administrator\AppData\Local\Temp\2\MktAnalytics_158_159_10_22
- We then show hidden folders and investigate deeper.

7. Inside of the .cache folder we find many screenshots and text files whenever the malware takes a picture of what the user is doing and sends it to its host.



8. It also leaves a txt file every time a screenshot is taken stating if the screenshot was successful as well as saying who designed the malware which was Redevil@2024

MARKETING SCREENSHOT CAPTURE LOG

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Capture ID: 15

Timestamp: 2026-05-03T18:11:01.871027

Process PID: 692

Username: Administrator

Hostname: Windows_Server

Storage Folder: C:\Users\ADMINI~1\AppData\Local\Temp\2\.MktAnalytics_158_159_10_22

Screenshot Success: True

MALWARE SIGNATURES:

Primary Signature: Redevil@2024

Tool Name: RedDevil_Marketing_Toolkit

Creator: Redevil@ Advanced Persistent Threat

Toolkit Version: RedDevil v3.2.1

9. We then run to TCPView and enter the process name to investigate any outbound connections which shows us that it is sending information to a remote IP address 158.159.10.22 over port 1443

MarketingTool.exe	5680	TCP	Syn Sent	10.0.35.189	49856	158.159.10.22	1443	5/2/2026 6:30:14 PM	MarketingTool.exe
MarketingTool.exe	5680	TCP	Syn Sent	10.0.35.189	49855	158.159.10.22	1443	5/2/2026 6:30:09 PM	MarketingTool.exe

10. So far we have the malware name, attacker ip, and the creator of the malware.

11. Next we head into autoruns to find the config data which is 7ab85

type	Data
REG_SZ	(value not set)
REG_SZ	7ab85
REG_SZ	"C:\Windows\System32\MarketingTool.e
REG_SZ	C:\Program Files\OpenVPN\bin\openvpr

12. The flag is marketingtool.exe:158.159.10.22:Redevil@2024:7ab85 thus finishing the lab

Recommendations

1. Block malicious IP address on network firewall to ensure no further communication with any other machines on the network.
2. Create a script that checks registry editor and runs a daily anti malware scan, mentioning all its findings in plain text documents.
3. Further cyber security training amongst end users such as phishing awareness and downloading unapproved software.